

Black Hole Instabilities and Cosmic Censorship

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Black Holes in $d = 4$

Kerr metric

$$ds^2 = - \left(1 - \frac{2Mr}{\Sigma} \right) dt^2 - \frac{4Mar \sin^2 \theta}{\Sigma} dt d\phi \\ + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \left(r^2 + a^2 + \frac{2Ma^2 r \sin^2 \theta}{\Sigma} \right) \sin^2 \theta d\phi^2$$



Black Holes in $d \geq 5$

Extension into higher dimensions not unique.

Schwarzschild-Tangherlini metric

$$ds^2 = -\left(1 - \frac{2M}{r^{d-3}}\right)dt^2 + \frac{dr^2}{1 - \frac{2M}{r^{d-3}}} + r^2 d\Omega_{d-2}^2$$

with horizon topology S^{d-1} .

Black String metric

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega_2^2 + dz^2$$

with horizon topology $S^2 \times S^1$ identifying $z \sim z + L$.

Quasinormal Modes

Perturbing the metric

$$g_{ab} \rightarrow g_{ab} + h_{ab}$$

gives rise to change in Ricci tensor

$$R_{ab} \rightarrow R_{ab} - \frac{1}{2} \Delta_L h_{ab} + O(h^2) ,$$

where

$$\Delta_L h_{ab} = \square h_{ab} + 2R_{acbd}h^{cd} - 2R^c_{(a}h_{b)c} - 2\nabla_{(a}\nabla^c h_{b)c} + \nabla_a\nabla_b h$$

so to first order, we require

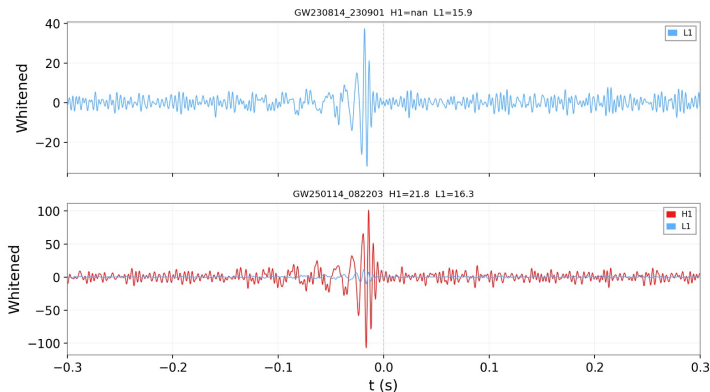
$$\Delta_L h_{ab} = 0 .$$

Quasinormal Modes

Look for solutions that oscillate in time

$$h_{ab} \sim e^{i\omega t}, \omega \in \mathbb{C}.$$

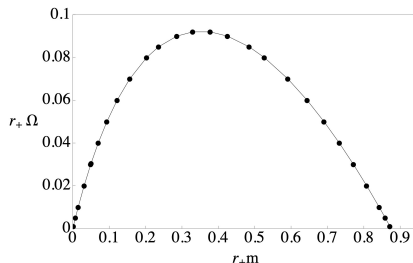
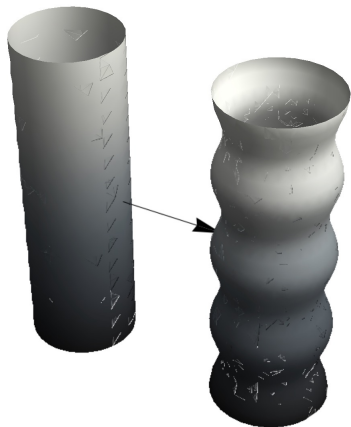
First two O4 events — Whitenened strain (zoomed ± 0.3 s around merger)



Gregory-Laflamme Instability

Look for perturbations of Black String ($S^2 \times S^1$) of form

$$h_{ab} = e^{imz} e^{\Omega t} H_{ab}(r) .$$



Thermodynamic Argument

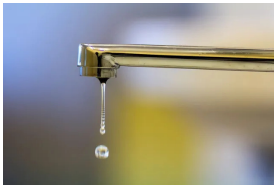
Can compute black hole entropy

$$S = \frac{A}{4}$$

for black hole and black string

$$S_{BH} = 4\pi M^2 \sqrt{\frac{8L}{27\pi M}}, \quad S_{BS} = 4\pi M^2$$

so as $L \rightarrow \infty$ the black hole is preferred.



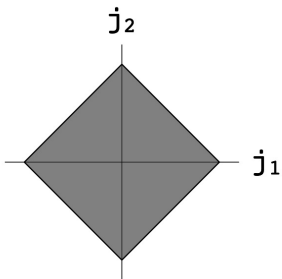
More Black Holes in $d \geq 5$

Can also generalise Kerr metric to Myers-Perry metric.

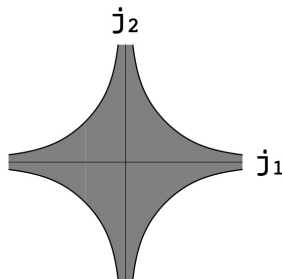
$$\Delta_4(r) = r_0^2 - 2Mr_0 + a^2 \rightarrow \Delta_d(r) = r_0^2 - 2Mr_0^{5-d} + a^2$$

Also possible to have up to $\lfloor \frac{d-1}{2} \rfloor$ angular momenta due to rank of $SO(d-1)$.

(a)

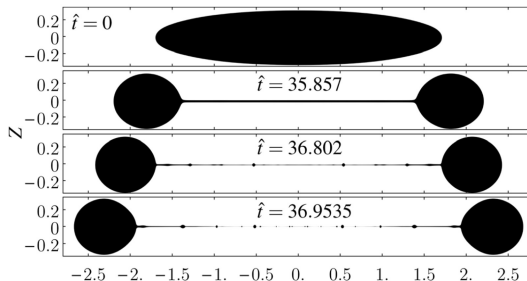
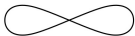
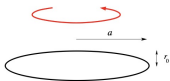


(b)

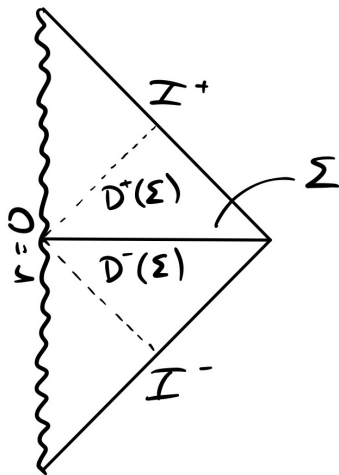


Ultraspinning Instability

High angular momentum causes deformation of event horizon (black pancake).



Cosmic Censorship



Black Branes

Can extend black strings to black branes

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega_2^2 + dz_1^2 + dz_2^2$$

and compactify on an oblique torus.

